Vanishing Gradient Problem

- The difficulty with vanishing gradients is that it makes it very difficult for the model to learn long-distance dependencies
- The effect of changes to the state in the *n*-th position before, is miniscule
  - Therefore, the hidden state in the *n*-th position before has no causal effect on the prediction *n* positions ahead
- This was addressed by a different RNN architecture called LSTM (Long Short Term Memory; Hochreiter and Schmidhuber, 1997)

Quantifying LSTM's Ability to Capture Long-distance Dependencies

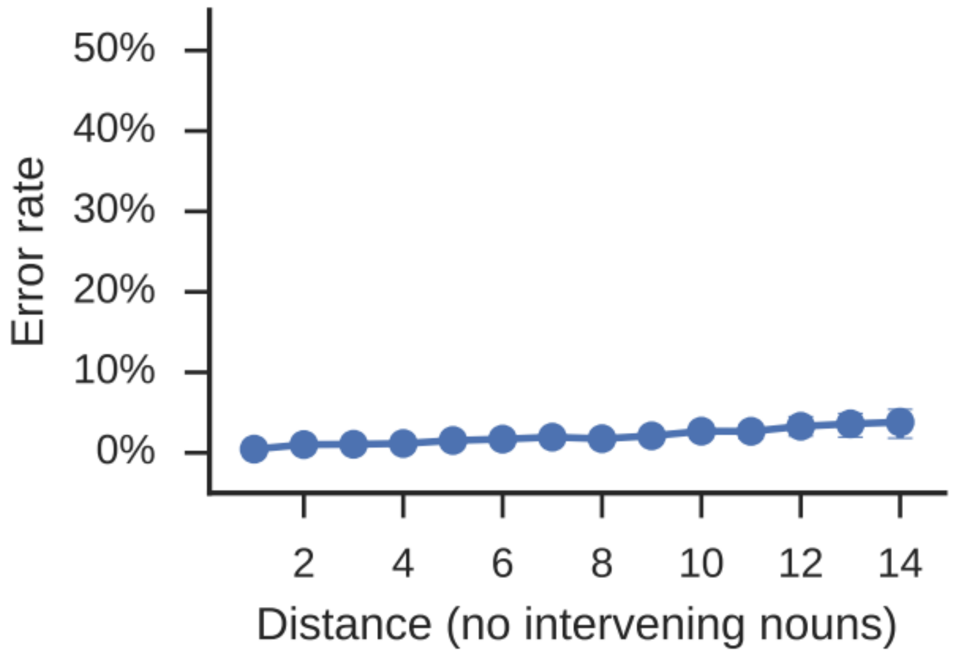
- Lizen et al. (2016) cast the problem as a subject-verb agreement problem
  - They tested mostly on English
  - They measured the effect both of distance and distractors on the error rate

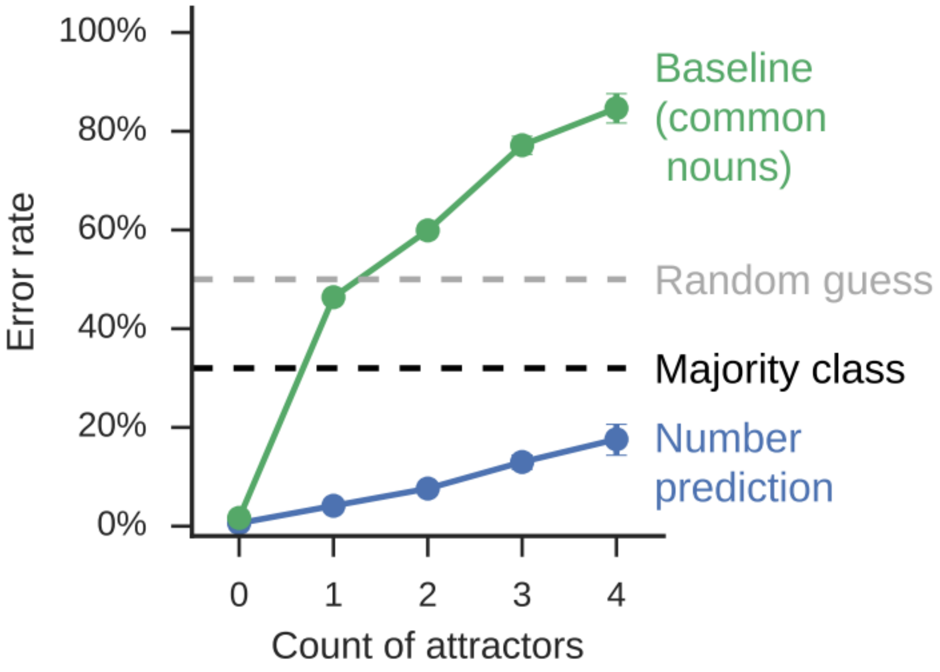
(9) The **roses** in the vase by the door **are** red.

(10) The **roses** in the vase by the *chairs* **are** red.

https://aclanthology.org/Q16-1037.pdf

Quantifying LSTM's Ability to Capture Long-distance Dependencies





The baseline is only exposed to the identity of the nouns, without the syntactic structure